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- 1.) Check quality with fastqc, but using falco (Galaxy Version 1.3.0+galaxy0) b/c faster
 - Default parameters
 - Run with transcriptome reads
 - Forward data
 - Fails: per tile sequence quality; Sequence duplication levels; Adapter Content
 - Warnings: Per base sequence content; Per sequence GC content; Overrepresented sequences
 - Reverse Data
 - Fails: Per tile sequence quality; adapter content
 - Warnings: Per base sequence content; Per sequence GC content; Sequence duplication levels; Overrepresented sequences

- 2.) Trim with fastp (Galaxy Version 1.3.3+galaxy0)
 - Set to paired collection
 - Gets rid of stuff that are fails for the quality control
 - Observe the results on HTML, those under filtering reports

- 3.) HISAT2 (Galaxy Version 2.2.2+galaxy0) - aligns transcriptome reads to the genome that is provided by Dr. Van Laar
 - Change in parameters:
 - Source for reference genome: genome from the history
 - Single or paired library? Set to paired-end dataset collection

Made an alignment file called BAM

Now, what actual transcripts do we have?

Step4.) StringTie ((Galaxy Version 2.2.3+galaxy0) - assembles short reads into actual transcripts, predicts full length transcripts

- Make sure input option parameter is set to Short Reads

Created 6 files of transcripts

- Change parameter to reference genome to history
 - Burkholderia.gff

Step 4.) StringTieMerge (Galaxy Version 2.2.3+galaxy0) - instead of 6 files, puts them all together

- Use dataset collection and use the StringTie file
- Use reference as burkholderia.gff

- step 5.) featureCounts (Galaxy Version 2.1.1+galaxy0)

- Set alignment file to dataset collection and use HISAT file
- Set gene annotation file as GFF/GTF from history and use latest stringtiemerge
- Rewrite GFF gene identifier as transcript_id

- Does the input have read pairs? Set to Yes, paired-end and count them as 1 single fragment
- Next step: Deseq2 ([Galaxy Version 2.11.40.8+galaxy2](#)) - function? Takes all 6 conditions and sees if it is more represented in one condition or the other, gives statistical significance
 - Make sure to rename to Counts under featurecounts for easier setup
- Factor: set to Gene_Type
- 2 factor levels. Factor level 1 set as WildType; Factor level 2 set as Mutant
 - Rename the counts as wildtype or mutants
 - Tells if gene was up or down regulated
 - Under Advanced options:
 - Use beta priors? Set to yes
 - Under Output options: generate plots and output normalised counts
 - Click “include hidden” within history and locate the featurecount and find the Counts, rename as plate/BrothReplicate#
 - Put these in using the multiple dataset options, search with the ellipses
- From the result file you get a p-value, can determine null hypothesis
 - For the gene, there is no change in gene expression between plate and broth: null hypothesis
 - If there are changes between the two growth media, you can infer significant difference, but need this p-value to make a statistical backing for your reasoning
 - Less than 0.05, it is statistically significant

Step 6.) **filter data on any column using simple expressions**

- Use on DEseq2 file
- With following condition - $c7 < 0.05$
 - Column 7 is the p-adj, run it on this one instead of the raw p-value column
- Run again with the following condition - $\text{abs}(c3) > 1$
 - Column 3 is the $\log_2(\text{FC})$, run on this
 - Only do this after the previous filtering is finished
- From the plots we download here, we learn about whether expression change between the two growth media/conditions is statistically significant enough to reject the null hypothesis

Step 7.) filter on dataset with following condition:

- $\text{abs}(c3) > 1$
 - Cut it down to 626 lines from 1010 lines

Step 8.) were gonna add a header now through a new Excel file

- Put the following in: GeneID Base mean $\log_2(\text{FC})$ StdErr Wald-Stats P-value P-adj

Step 9.) concatenate tool

- Insert the most recent filter on dataset file

Now our data file has proper headers!

- Rename the file to Differential Gene Expression of Mutant and Wildtype

Step 10.) volcano plot ([Galaxy Version 4.0.2+galaxy0](#)) - represents p-values

- Change the respective columns into the following: 7, 6, 3, 1